

Alex Rivera — AI/ML Research Engineer

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Executive Summary

Senior AI & Systems Engineer with 3+ years experience building local LLM inference engines, RAG pipelines, and deterministic agent orchestration frameworks. Passionate about privacy-first intelligence, low-latency quantized inference, and verifiable autonomous systems.

Technical Skills

- **Core Languages:** Python (Advanced), Rust (Intermediate), C++ (Prior experience), TypeScript
- **Machine Learning & Frameworks:** PyTorch, Hugging Face Transformers, vLLM, Ollama, ONNX Runtime, SentenceTransformers
- **Retrieval & RAG:** ChromaDB, LanceDB, FAISS, Hybrid Sparse-Dense Retrieval, Reciprocal Rank Fusion (RRF)
- **Agent Systems & Safety:** Structured JSON schema generation, Tool-use safety rails, Human-in-the-loop permission gateways, Deterministic sandboxing
- **Backend & Cloud:** FastAPI, Uvicorn, Docker, SQLite, AsyncIO, WebSockets, Linux Systems

Experience

AI Systems Engineer | VeriLocal Systems (2024 – Present)

- Designed an on-device privacy-preserving RAG engine indexing over 50,000 internal documents with sub-80ms semantic retrieval latency.
- Implemented human-gated tool execution protocol reducing unauthorized system modifications to 0%.
- Engineered continuous verification hooks that validate state integrity post-action on local disk.

Machine Learning Intern | Cognition Labs (Summer 2023)

- Built automated evaluation benchmarks for local quantized models (Qwen, Llama-3, Mistral) measuring JSON schema adherence and tool call precision.
- Fine-tuned 7B parameter SLMs for structured reasoning tasks achieving 94.2% valid tool-dispatch syntax.

Key Projects

Project Alpha: Local Vector Memory Engine (2024)

- Embedded zero-dependency hybrid vector search engine in Python and PyTorch.

- Implemented streaming token generation with live tool dispatch checkpoints and audit logging.

SecureAgent Sandbox (2023)

- Open-source permissioned agent runtime with explicit write protection, path-traversal prevention, and append-only cryptographic audit logging.
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Education

B.S. in Computer Science (Artificial Intelligence Track)

Stanford University — GPA: 3.92 / 4.0 (Graduated 2024)

Relevant Coursework: CS224N (NLP with Deep Learning), CS231N (Computer Vision), CS229 (Machine Learning), CS110 (Principles of Computer Systems)